

Computer Vision(YOLOv8) & Restricted Zone Monitor --- Jyotirmoy Bhunia

What is Spatial Intelligence?

- Spatial intelligence refers to an AI system's ability to perceive, reason about, and interact with the three-dimensional (3D) world around it. Spatial intelligence involves reasoning about depth, geometry, movement, and the relationships between objects in a dynamic environment
- While closely related, it is helpful to distinguish spatial intelligence vs. computer vision →
 1. **Computer Vision** is the broader field focused on deriving meaningful information from digital images, videos, and other visual inputs. It includes tasks like classification or basic 2D detection.
 2. **Spatial Intelligence** is a specialized subset or evolution of computer vision that specifically adds the dimension of *space* and *physics*. It moves from "What is this object?" (Vision) to "Where is this object, how is it oriented, and how can I interact with it?" (Spatial Intelligence).

Applications in Drone Technology

- **SLAM (Simultaneous Localization and Mapping):** This allows a device, such as a robot or drone, to map an unknown environment while keeping track of its own location within it. Modern approaches often integrate visual SLAM with deep learning to improve robustness in changing lighting conditions.
- SLAM helps in real time mapping , location tracking, build & update the map of surroundings.

What is Telemetry?

- Telemetry is the automated collection and transmission of data and measurements from distributed or remote sources to a central system for [monitoring](#), analysis and [resource optimization](#).
- In industries that rely on physical assets, such as [agriculture](#), [utilities](#) and [transportation](#), organizations use telemetry to capture measurements such as temperature, air pressure, motion and light.

EXAMPLE OF TELEMETRY

- Telemetry is one of the latest ways to track software progress. It helps developers stay aware of the software's performance and notifies them if a problem occurs with the application.
- Software development is a great way to learn which features end-users prefer and which require modification. This system also notifies the developers if there is a problem with the application, like a bug. It gives the developers enough time and data to rectify the problem and introduce an even better version in the market. It is an important step in ensuring the end-user gets the best application version

How Telemetry works for application monitoring



Specifying metrics

Developers identify key metrics to track critical aspects of application performance.



Data transmission

Data is securely transmitted from the application to a remote server.



Data processing

Raw data is processed into insights using formulas and algorithms.



Data analysis

Analyzed data reveals performance issues, guiding optimizations.

What is State-of-the-art (SOTA) ?

- State-of-the-art (SOTA) AI models are the most advanced and innovative models currently available. They represent the highest level of achievement in a specific area of AI research, often setting new standards for performance and capability.

Ex: 1. NLP -> GPT-4 (OpenAI)

2. CV -> Vision Transformers (ViT) (Google)

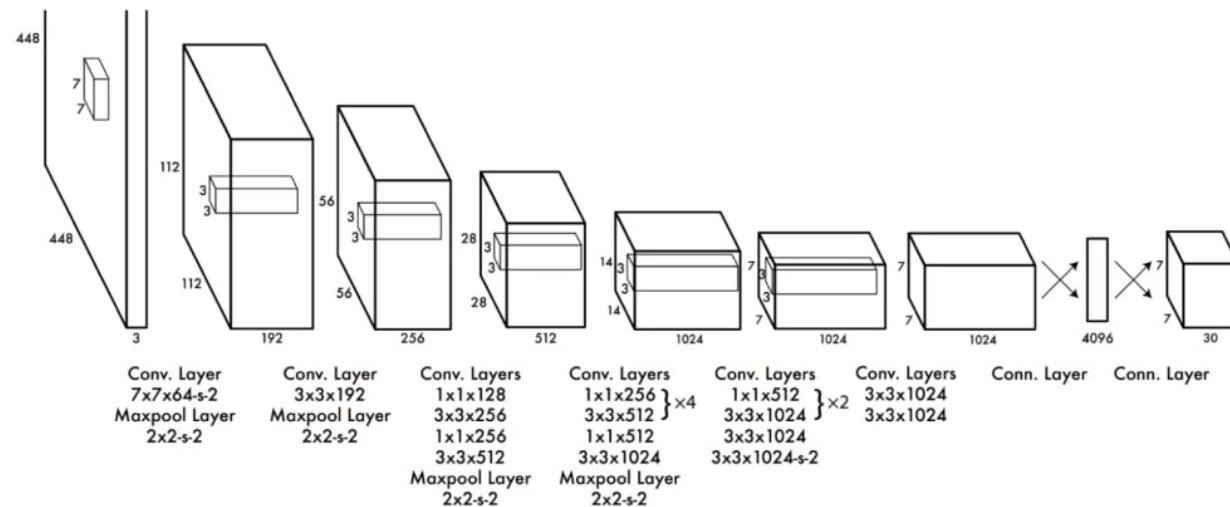
What is YOLO?

- You Only Look Once (YOLO) proposes using an end-to-end [neural network](#) that makes predictions of bounding boxes and class probabilities all at once.

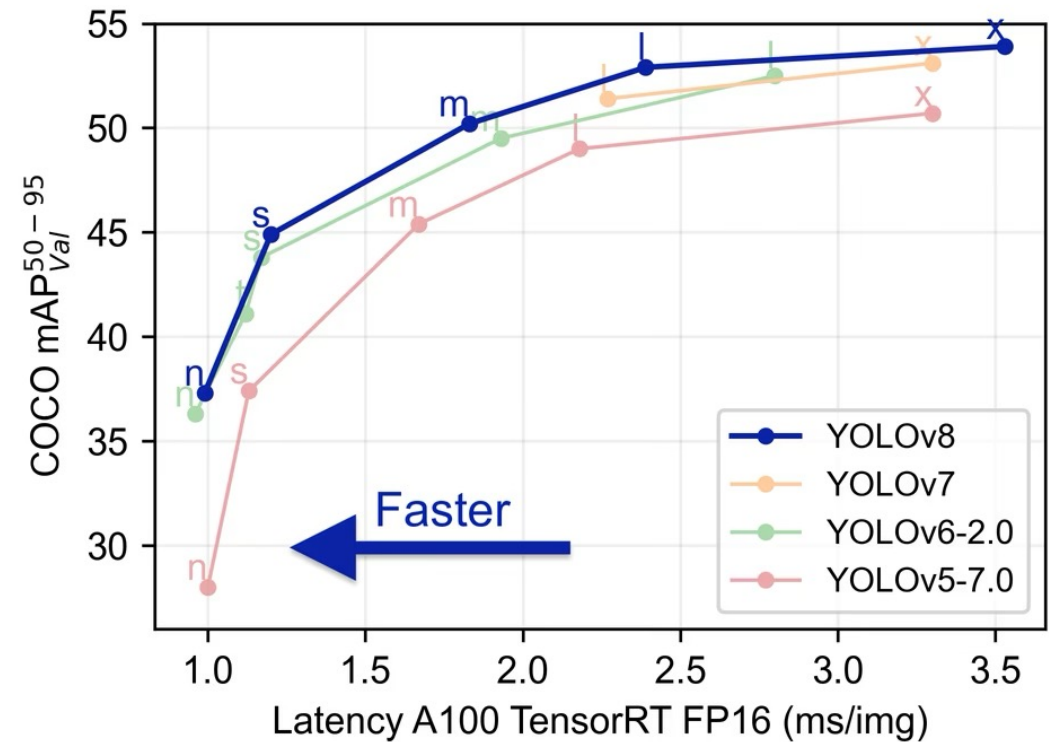
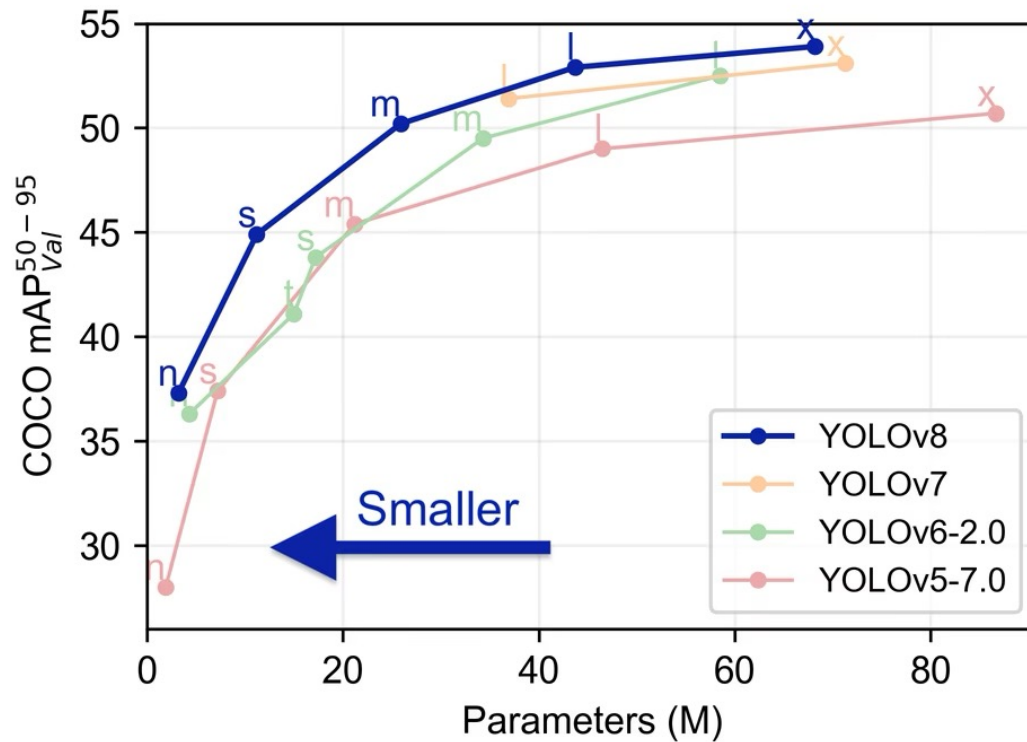


Architecture of YOLO

- The [YOLO algorithm](#) takes an image as input and then uses a simple deep convolutional neural network to detect objects in the image. The architecture of the CNN model that forms the backbone of YOLO is shown below.



The Architecture. Our detection network has 24 convolutional layers followed by 2 fully connected layers. Alternating 1×1 convolutional layers reduce the features space from preceding layers. We pretrain the convolutional layers on the ImageNet classification task at half the resolution (224×224 input image) and then double the resolution for detection.

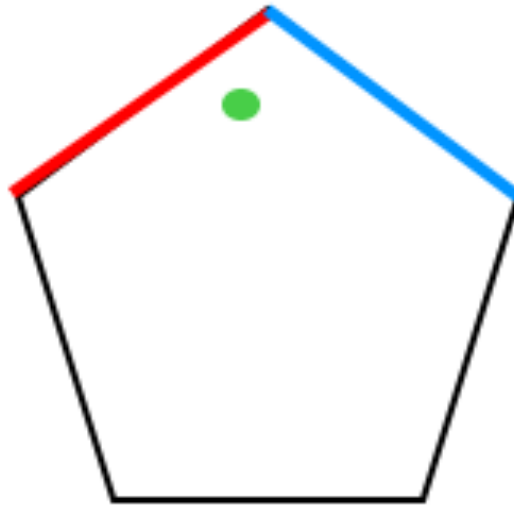


Model Comparison

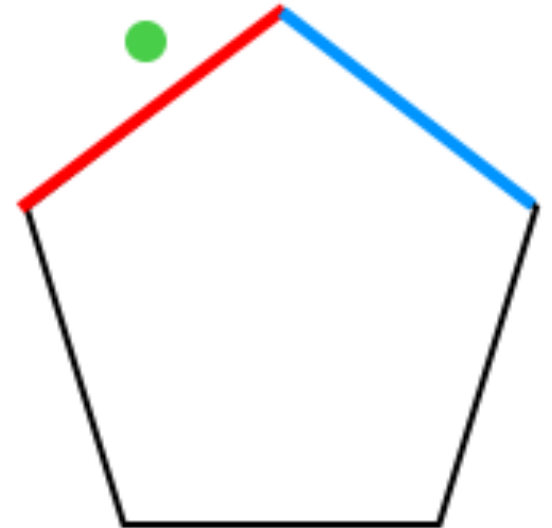
- Graph-1: At ~25M parameters, YOLOv8-m scores ~50.2 mAP, while YOLOv6 and v5 score noticeably lower at the same size (small model with great accuracy)
- Graph-2: At ~1.2ms latency, YOLOv8-s hits ~44.9 mAP, while YOLOv5 at the same speed scores much lower (for the same inference speed v8 runs faster)

What is Point-in-Polygon-logic ?

- Point-in-Polygon (PIP) logic is a computational geometry method used to determine if a specific point lies inside, outside, or on the boundary of a defined polygon



Inside example



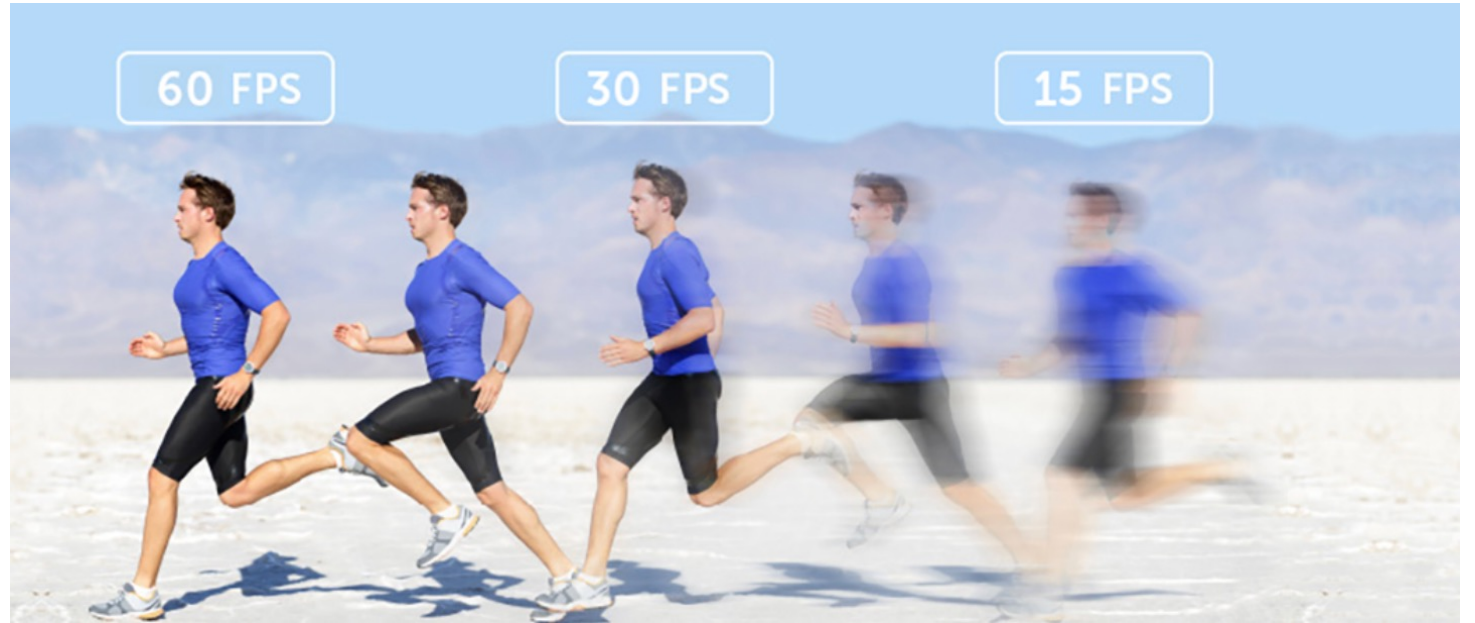
outside example

What is latency(lag) in video?

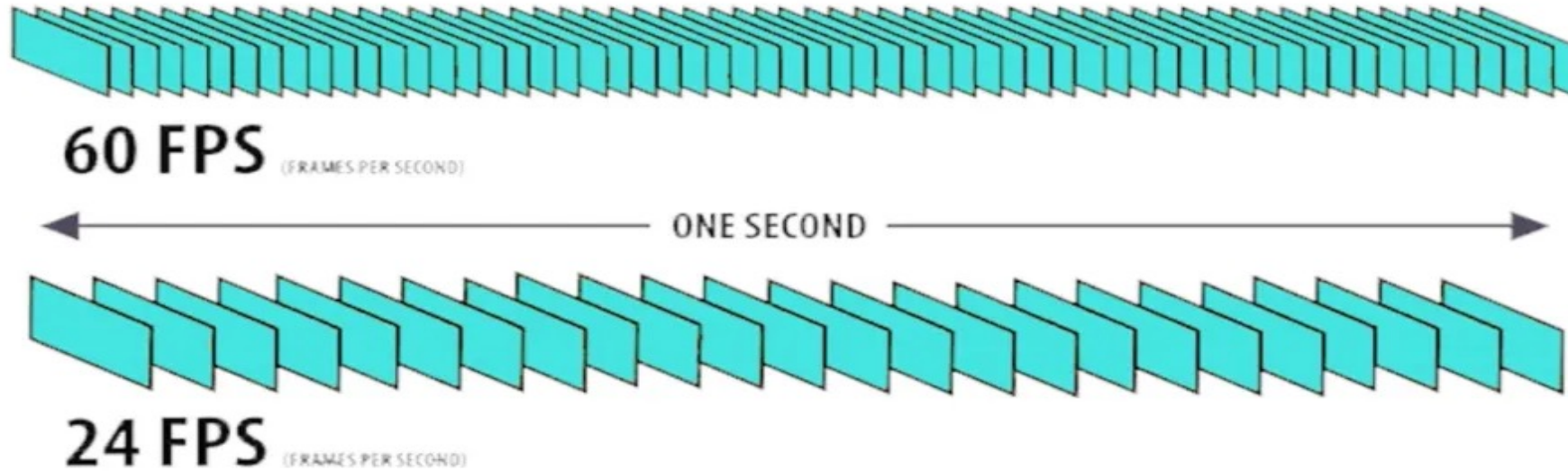
- The time it takes for a video signal to travel from the source to the viewer. The lower the latency, the better the video quality and viewer experience.
- When latency is high, it can cause problems like delays and buffering.

What is FPS(Frames Per Second)?

- Frames per second (FPS) or frame rate indicates the number of individual frames that are processed or displayed consecutively in a second
- FPS measures the number of still images, or frames, displayed in one second of a video or animation clip.
- The higher the number of frames in a video, the smoother it looks. On the other hand, a lower frame rate makes the video appear choppy or stuttered.

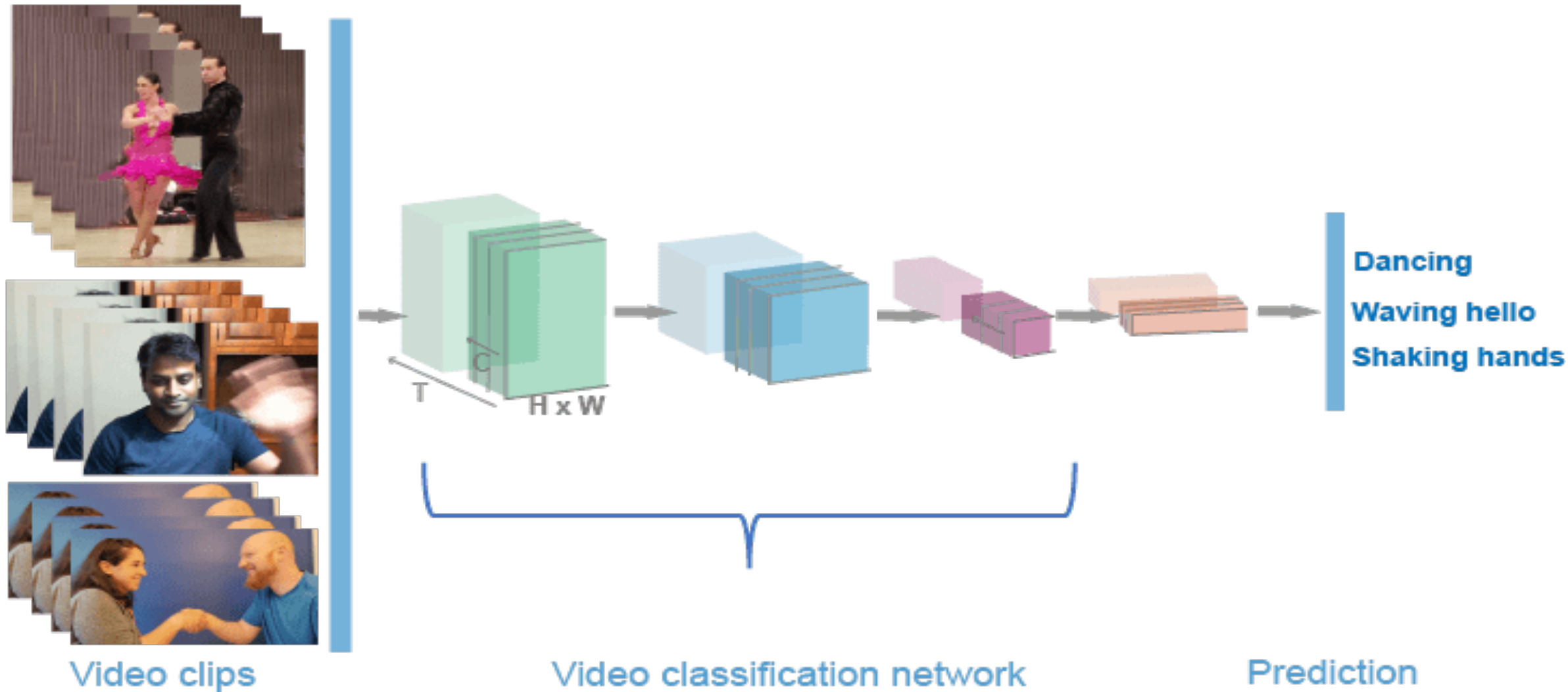


Frames Per Second: FPS

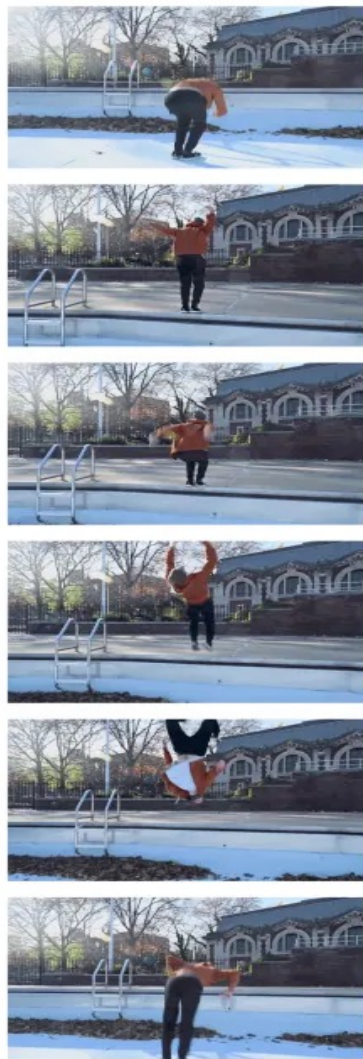


- $FPS = \text{Number of Frames} / \text{Duration in Seconds}$

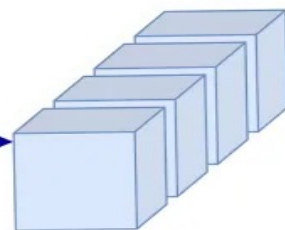
Role of FPS in Vision Models



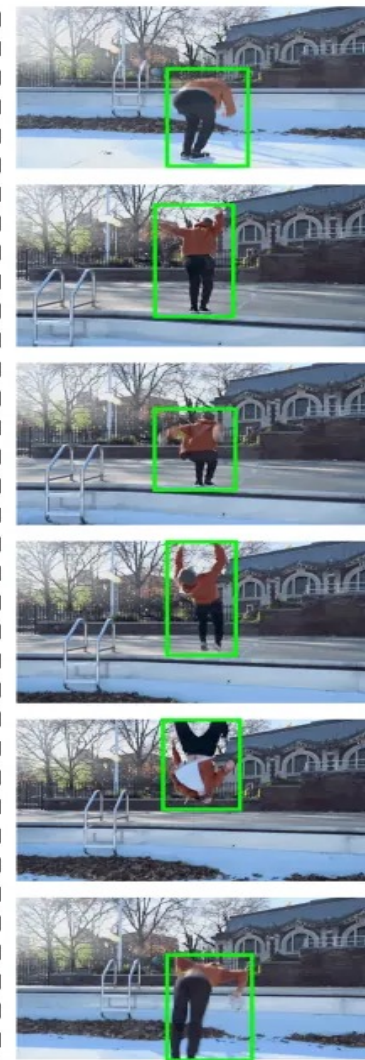
Input Video



3D Convolutions



Object Detection & Tracking



Activity Classification

